NGUYEN QUANG TRUNG

AI Engineer

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OBJECTIVE

AI Engineer with a strong background in deep learning, computer vision, and natural language processing. Experienced in designing and implementing AI-driven solutions across multiple domains, including healthcare, finance. Proficient in Python and AI frameworks such as TensorFlow, PyTorch, and Scikit-learn. Passionate about leveraging AI to drive innovation and solve real-world problems. Seeking opportunities to contribute expertise in artificial intelligence and machine learning.

EDUCATION

FPT School of Business & Technology 2024 - Present

Master of Software Engineering - Artificial Intelligence

GPA: 8.6/10

FPT University HCMC 2019 - 2024

Bachelor of Information Technology

Major: Artificial Intelligence
GPA: 7.0/10

SKILLS

Programming Languages	Python, Java
Deep Learning & Machine Learning	TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers
Data Analysis	Pandas, Numpy, Matplotlib, Seaborn, OpenCV
Databases	MySQL, SQLite
Software & Tools	Git, Docker, Jupyter Notebook, Streamlit
Soft Skills	Problem-solving, teamwork, critical thinking, effective communication
Languages	English (Intermediate proficient in reading documentation and technical papers)

CERTIFICATIONS

Natural Language Processing (Coursera)	2023
Google Project Management (Coursera)	2023
Big Data (Coursera)	2024
Data Mining Foundations and Practice (Coursera)	2025
Agile Development (Coursera)	2025

PROJECTS

AI-Based Early Warning System for Student Mental Health -

- **Built a machine learning system to detect early signs of depression and anxiety in students using PHQ-9 and GAD-7 survey data.**
- **Preprocessed question-level responses and response times; mapped features to clinical symptoms for interpretability.**
- **Trained classification models (Random Forest, XGBoost) to predict mental health risk levels (none, mild, moderate, severe).**
- **Developed a prototype Streamlit dashboard to visualize results and support teachers/psychologists in early intervention. Github:**

Tech Stack: Python, Pandas, NumPy, Scikit-learn, SHAP, Matplotlib/Seaborn, Streamlit
Github:

Movie Recommendation System -

- **Built an end-to-end pipeline on Google Colab using MovieLens 20M dataset**
- **Preprocessed data: type normalization, timestamp to datetime, release year & genres extraction, deduped repeated ratings; filtered low-activity users/items; per-user chronological split to avoid temporal leakage.**
- **Performed EDA to characterize sparsity and long-tail behavior, temporal trends, and genre distributions to guide modeling and evaluation.**
- **Implemented lightweight baselines and set up evaluation with RMSE/MAE and ranking metrics using offline negative sampling.**

Tech Stack: Python, Pandas, NumPy, Matplotlib
Github:

Voice Separation and Gender Recognition System -

- **Developed a deep learning system to separate overlapping voices and classify speaker gender from audio recordings.**
- **Implemented a time-domain speech separation model using a Temporal Convolutional Network (TCN) architecture trained on a Vietnamese dataset of audiobook voice mixtures.**
- **Engineered a gender recognition model using a 5-layer dense neural network trained on balanced male/female samples from Mozilla Common Voice, with 128-dim Mel spectrogram features.**
- **Built a full-stack Streamlit web app for real-time audio recording/upload, waveform & spectrogram visualization, speech separation, and gender prediction with confidence scores.**

Tech Stack: Python, PyTorch, TensorFlow/Keras, Streamlit, Librosa, Torchaudio, PyAudio, NumPy, Matplotlib
Github:

Cancer Segmentation in CT Scans (Capstone Project) -

- **Developed a deep learning system to segment cancerous organs (pancreas) from CT scans.**
- **Applied MONAI transforms and trained models on the PANORAMA dataset using Dice loss, Adam optimizer, and AUC-ROC for evaluation.**
- **Implemented and evaluated 2D U-Net with interchangeable encoders (ResNet50, VGG19, MobileNetV2), as well as 3D U-Net and transformer-based Swin-UNETR for volumetric segmentation.**
- **Built a full-stack Streamlit web demo app for interactive CT scan upload, slice-wise visualization, and real-time segmentation with different models.**

Tech Stack: Python, PyTorch, MONAI, Scikit-learn, Matplotlib, NiBabel, Streamlit, SimpleITK
Github: